

Study of the Effect of Pseudo-Sine Wave as a LO on the Linearity Performance of the RF Mixer

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Abstract—This paper provides the study of the effect of using Pseudo-Sine wave as a local oscillator (LO) on the linearity performance of the RF mixer instead of sine wave. The frequency range of operation is chosen to target Internet of Things (IoT) applications. In a mixer, when the incoming RF signal is convolved with the sine wave Local Oscillator (LO), it leads to down-conversion of the higher order components of RF signal within the desired frequency band resulting in distortions and degradation of Spurious Free Dynamic Range (SFDR) of the receiver, to pull off this issue a harmonic free pseudo-sine VCO is realised using a ring oscillator. The circuit is implemented in 180nm CMOS process in UMC library using cadence virtuoso tool.

Keywords—IOT, RF, Pseudo-sine VCO, Oscillator, VCO, CMOS Ring Oscillator.

I. INTRODUCTION

In this era, Internet-of-Things (IoT) is the future of wireless personal communication system which connects electronic devices and gadgets [3]. Electronics industry is focusing on the consumer appliances with their devices to upgrade in terms of wireless connectivity [1-2]. This enables the consumer to access their smart television, fridge, washing machine and other wirelessly connected smart devices from anywhere on the planet. Clock generation is an important aspect as the functionality of the modern digital devices completely depends on the clock. Local oscillators (LO), therefore, have a significant role and the circuit implementation of the LO is critical for the wireless system to achieve proper dynamic range (DR) of the wireless system. Due to the demand of high-performance in high power computing, the IoT devices are required to operate with high dynamic range.

The dynamic range of RF wideband multi-standard receiver is limited by its linearity performance. Distortion free input ascertains the improvement in the conversion gain, sensitivity and the Noise Figure(NF) for wideband applications.

The harmonics present at the LO input to wideband down-conversion RF mixer, the odd harmonics are also down converted to baseband and thus generating Intermediate

Frequency (IF) signals which contains spurious signals, and hence degrading the overall Spurious Free Dynamic Range (SFDR) of the receiver. To accomplish better signal quality with minimum distortions at the receiver, the RF mixer must and should be designed to have a wide Dynamic Range(DR) and linearity to process the signal from Low Noise Amplifier (LNA). The LO signal with appropriate phase shifts and opposing symbols for the higher order harmonics coefficients (i.e. third and fifth order) helps reducing their significance and thus improving the linearity of the mixer. To reduce the significance of third- and fifth- order harmonics, pseudo-sine wave can be used instead of the conventional sine wave as LO. In this paper, the effect of the pseudo-sine wave on the linearity performance of the mixer is studied. In this work, the mathematical equations governing the pseudo-sine wave is derived in section – II, and the circuit level implementation of the pseudo-sine LO is discussed in section – III. The designed pseudo-sine LO is applied to a standard Gilbert Mixer and its performance is evaluated in section – IV, and the work is concluded in section – V.

II. PSEUDO-SINE SIGNAL MATHEMATICS

Passive resistive interpolation [5] scheme is power and area efficient, and can be well-matched in layout via upsizing the physical dimension of the resistors (e.g., poly-silicon). Here, a 3-path passive resistive interpolation methodology is included after the inverter-based buffers [4,7] for I/Q generation. It is basically used for RF applications such as cellular mobile transceiver systems by improving the linearity of the down-converter. In hard switching mixer, when the RF signal coming from the receiver gets multiplied with the LO signal which is a square type signal, then it leads to the down conversion of higher (i.e. 3rd and 5th) order component of the RF signal located at the harmonics of LO to the desired IF frequency band [5-9], which causes distortions in the output & degrades the linearity. In order to overthrow this issue, instead of using normal LO we may use a harmonic free pseudo-sine LO which is synthesized from a square LO as shown in the figure 1.

In this study, a Voltage controlled Oscillator (VCO) is implemented instead of simple LO, so that the frequency

2019 International Conference on Vision Towards Emerging Trends in Communication and Networking (ViTECoN) can be varied depending upon our requirement. The mathematical expression for pseudo-sine wave is derived using Fourier series expansion as follows.

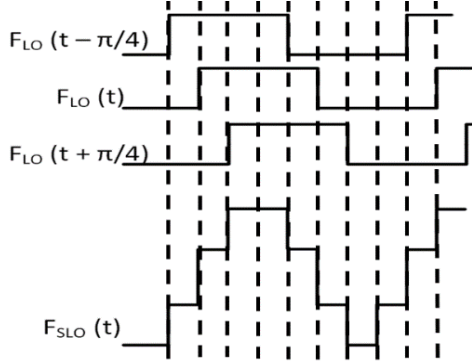


Fig. 1. Synthesis of Pseudo-sine signal using square waves [7].

The Fourier series expansion of square wave $F_{LO}(\omega t)$ is given by:

$$F_{LO}(\omega t) = \frac{2}{\pi} \left\{ \cos \omega t - \frac{1}{3} \cos 3\omega t + \frac{1}{5} \cos 5\omega t - \dots \right\} \quad (1)$$

And $F_{LO}(\omega t - \frac{\pi}{4})$ & $F_{LO}(\omega t + \frac{\pi}{4})$ is calculated as:

$$F_{LO}(\omega t - \frac{\pi}{4}) = \frac{2}{\pi} \left\{ \cos(\omega t - \frac{\pi}{4}) - \frac{1}{3} \cos 3(\omega t - \frac{\pi}{4}) + \frac{1}{5} \cos 5(\omega t - \frac{\pi}{4}) - \dots \right\} \quad (2)$$

$$F_{LO}(\omega t - \frac{\pi}{4}) = \frac{2}{\pi} \left\{ \left(\cos \omega t \cdot \cos \frac{\pi}{4} + \sin \omega t \cdot \sin \frac{\pi}{4} \right) - \frac{1}{3} \left(\cos 3\omega t \cdot \cos \frac{3\pi}{4} + \sin 3\omega t \cdot \sin \frac{3\pi}{4} \right) + \frac{1}{5} \left(\cos 5\omega t \cdot \cos \frac{5\pi}{4} + \sin 5\omega t \cdot \sin \frac{5\pi}{4} \right) - \dots \right\} \quad (3)$$

$$F_{LO}(\omega t - \frac{\pi}{4}) = \frac{2}{\pi} \left\{ \left(\frac{\cos \omega t}{\sqrt{2}} + \frac{\sin \omega t}{\sqrt{2}} \right) - \frac{1}{3} \left(-\frac{\cos 3\omega t}{\sqrt{2}} + \frac{\sin 3\omega t}{\sqrt{2}} \right) + \frac{1}{5} \left(-\frac{\cos 5\omega t}{\sqrt{2}} - \frac{\sin 5\omega t}{\sqrt{2}} \right) - \dots \right\} \quad (4)$$

$$F_{LO}(\omega t - \frac{\pi}{4}) = \frac{\sqrt{2}}{\pi} \left\{ (\cos \omega t + \sin \omega t) + \frac{1}{3} (\cos 3\omega t - \sin 3\omega t) - \frac{1}{5} (\cos 5\omega t + \sin 5\omega t) - \dots \right\} \quad (5)$$

$$F_{LO}(\omega t + \frac{\pi}{4}) = \frac{2}{\pi} \left\{ \cos(\omega t + \frac{\pi}{4}) - \frac{1}{3} \cos 3(\omega t + \frac{\pi}{4}) + \frac{1}{5} \cos 5(\omega t + \frac{\pi}{4}) - \dots \right\} \quad (6)$$

$$F_{LO}(\omega t + \frac{\pi}{4}) = \frac{2}{\pi} \left\{ \left(\cos \omega t \cdot \cos \frac{\pi}{4} - \sin \omega t \cdot \sin \frac{\pi}{4} \right) - \frac{1}{3} \left(\cos 3\omega t \cdot \cos \frac{3\pi}{4} - \sin 3\omega t \cdot \sin \frac{3\pi}{4} \right) + \frac{1}{5} \left(\cos 5\omega t \cdot \cos \frac{5\pi}{4} - \sin 5\omega t \cdot \sin \frac{5\pi}{4} \right) - \dots \right\} \quad (7)$$

$$F_{LO}(\omega t + \frac{\pi}{4}) = \frac{2}{\pi} \left\{ \left(\frac{\cos \omega t}{\sqrt{2}} - \frac{\sin \omega t}{\sqrt{2}} \right) - \frac{1}{3} \left(-\frac{\cos 3\omega t}{\sqrt{2}} - \frac{\sin 3\omega t}{\sqrt{2}} \right) - \dots \right\} \quad (8)$$

$$\frac{\sin 3\omega t}{\sqrt{2}} + \frac{1}{5} \left(-\frac{\cos 5\omega t}{\sqrt{2}} + \frac{\sin 5\omega t}{\sqrt{2}} \right) - \dots \}$$

$$F_{LO}(\omega t + \frac{\pi}{4}) = \frac{\sqrt{2}}{\pi} \left\{ (\cos \omega t - \sin \omega t) + \frac{1}{3} (\cos 3\omega t + \sin 3\omega t) - \frac{1}{5} (\cos 5\omega t - \sin 5\omega t) - \dots \right\} \quad (9)$$

When we sum these equations (1), (5) & (9), we get

$$F_{SLO}(\omega t) = F_{LO}(\omega t - \frac{\pi}{4}) + F_{LO}(\omega t) + F_{LO}(\omega t + \frac{\pi}{4}) \quad (10)$$

$$F_{SLO}(\omega t) = \frac{\sqrt{2}}{\pi} \left\{ (\cos \omega t + \sin \omega t) + \frac{1}{3} (\cos 3\omega t - \sin 3\omega t) - \frac{1}{5} (\cos 5\omega t + \sin 5\omega t) - \dots \right\} + \frac{2}{\pi} \left\{ \cos \omega t - \frac{1}{3} \cos 3\omega t + \frac{1}{5} \cos 5\omega t - \dots \right\} + \frac{\sqrt{2}}{\pi} \left\{ (\cos \omega t - \sin \omega t) + \frac{1}{3} (\cos 3\omega t + \sin 3\omega t) - \frac{1}{5} (\cos 5\omega t - \sin 5\omega t) - \dots \right\} \quad (11)$$

$$F_{SLO}(\omega t) = 2 \cdot \frac{\sqrt{2}}{\pi} \left\{ \cos \omega t + \frac{1}{3} \cos 3\omega t - \frac{1}{5} \cos 5\omega t - \dots \right\} + \frac{2}{\pi} \left\{ \cos \omega t - \frac{1}{3} \cos 3\omega t + \frac{1}{5} \cos 5\omega t - \dots \right\} \quad (12)$$

Here the coefficients of 3rd & 5th order are equal and opposite except to their magnitude, so we do need to scale the $F_{LO}(\omega t)$ by $\sqrt{2}$. So the new equation to calculate the final output is:

$$F_{SLO}(\omega t) = F_{LO}(\omega t - \frac{\pi}{4}) + \sqrt{2} \cdot F_{LO}(\omega t) + F_{LO}(\omega t + \frac{\pi}{4}) \quad (13)$$

$$F_{SLO}(\omega t) = 2 \cdot \frac{\sqrt{2}}{\pi} \left\{ \cos \omega t + \frac{1}{3} \cos 3\omega t - \frac{1}{5} \cos 5\omega t - \dots \right\} + 2 \cdot \frac{\sqrt{2}}{\pi} \left\{ \cos \omega t - \frac{1}{3} \cos 3\omega t + \frac{1}{5} \cos 5\omega t - \dots \right\} \quad (14)$$

$$F_{SLO}(\omega t) = \frac{4\sqrt{2}}{\pi} \cos \omega t \quad (15)$$

The above derivation (1) – (15) gives a key idea about how the pseudo-sine LO signal i.e. $F_{SLO}(\omega t)$ is synthesized with the help of 3-phase shifted square signals with the assumption that there is no gain or phase mismatch [9]. In the next section, the circuit implementation of the pseudo-sine LO is discussed.

III. CIRCUITS DESIGN AND IMPLEMENTATION

The design of the Pseudo-sine VCO is implemented in cadence virtuoso by using ring VCO, with passive resistive interpolation scheme. The frequency tuning is done by V_{bias} input voltage. The Ring-LO is made up of series of odd number of inverters, and to provide enable (EN) control input, a NAND gate is used. To make it as a Ring-VCO, a NMOS as a current sink is used in each inverter stage and the control biased voltage (V_{bias}) is connected to it. The circuit schematic of Ring-VCO with enable input is shown in figure 2. Figure 3 shows the passive resistive interpolation scheme, which is

used to convert the phase shifted waves to the pseudo-sine wave as output at 2.4 GHz, shown in figure 4.

Fig. 3. Passive resistive interpolation scheme.

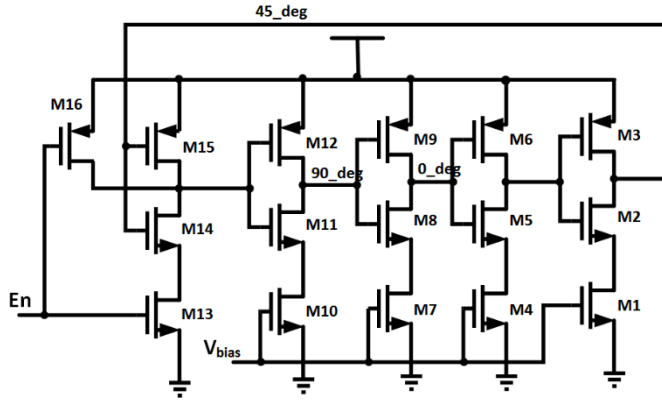


Fig. 2. Ring-VCO at 2.4 GHz

The phase noise (PN) performance of a VCO will alter the overall performance of the system in which the oscillator is used. Hence, the PN of the pseudo-sine LO is evaluated in figure 5. The implemented pseudo-sine LO achieves good PN performance of -81.52dBc/Hz and -101.25dBc/Hz at an offset of 1MHz and 5MHz respectively. The achieved PN performance is better than the required PN of -73.5 dBc/Hz [1,2].

The VCO tuning sensitivity is shown in figure 6. The VCO tuning sensitivity K is given as:

$$KVCO = \frac{\Delta F}{\Delta V} \quad (16)$$

where, ΔF is the change in frequency,

ΔV is the change in applied bias voltage

It can be seen from Fig. 6 that the frequency can be varied from 0 – 7.5 GHz for a bias voltage for 1.25 – 0.5 V.

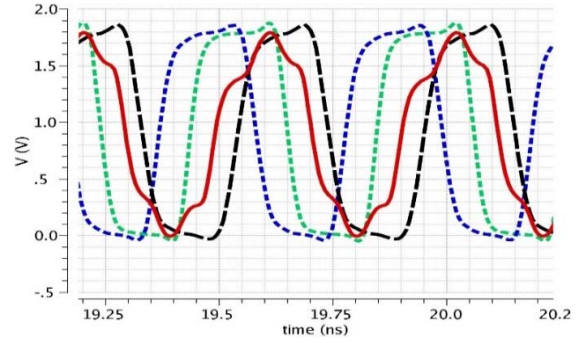
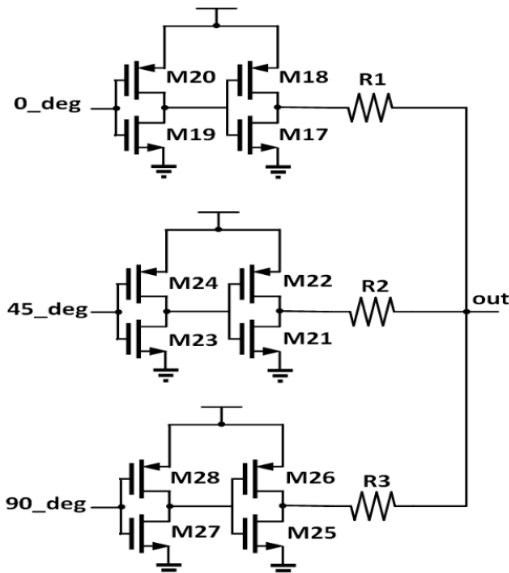


Fig. 4. output pseudo-sine waveform

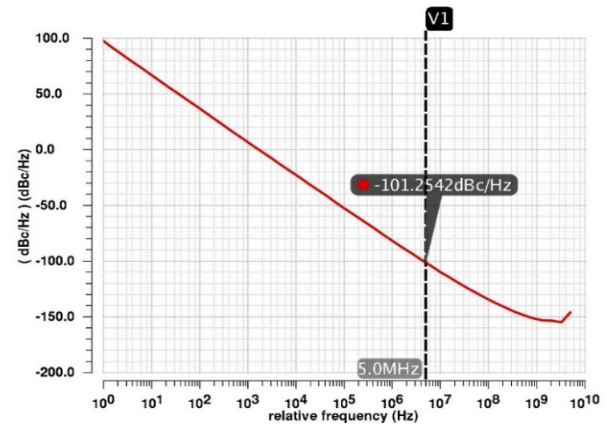
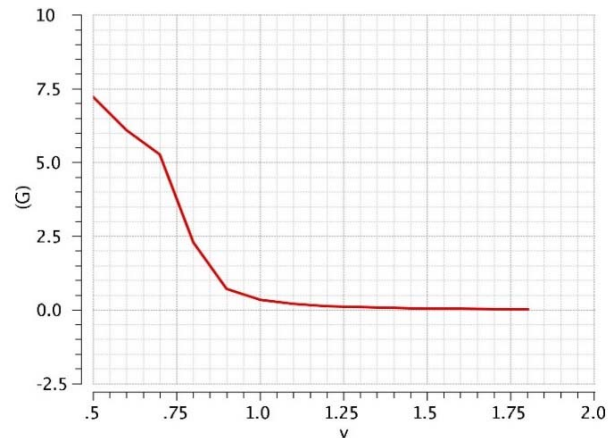


Fig. 5. Phase Noise of Pseudo-sine VCO

TABLE I. Phase Noise

Frequency	Phase Noise
1 MHz	-81.52 dBc/Hz
5 MHz	-101.25 dBc/Hz
10 MHz	-109.74 dBc/Hz
100 MHz	-134.24 dBc/Hz



IV. PERFORMANCE ANALYSIS OF IMPLEMENTED PSEUDO-SINE VCO ON THE GILBERT MIXER

Testing the RF Down-conversion performance using Pseudo-Sine and Sine wave by taking the active Gilbert Mixer stage shown in figure 7 below. The testing is done in UMC 180nm process.

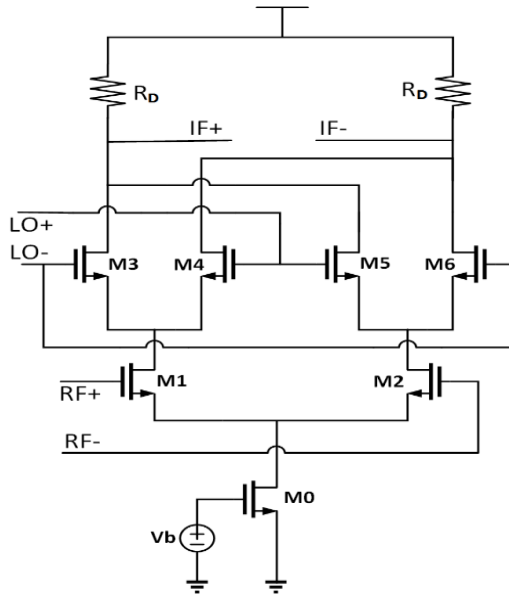
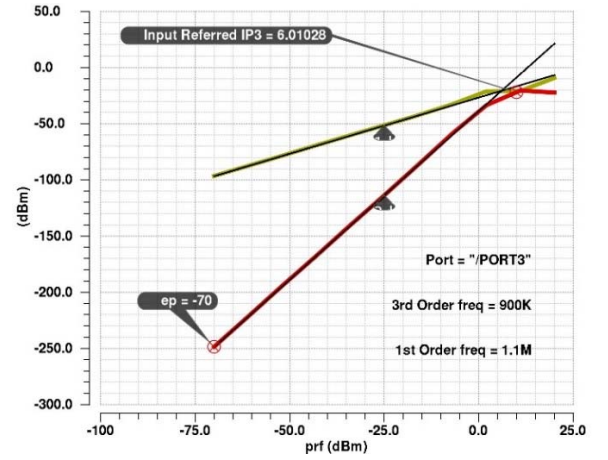
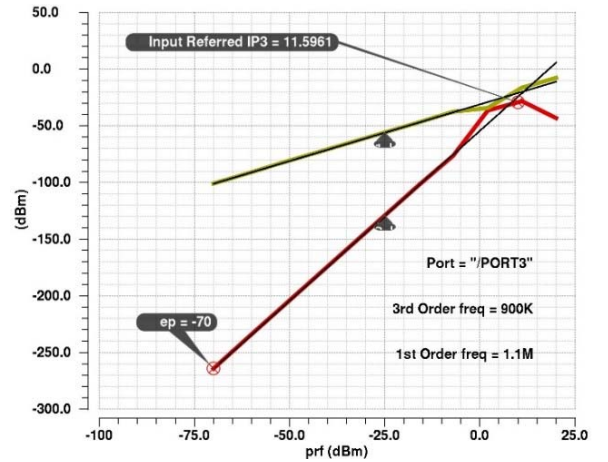


Fig. 7. Setup for testing Pseudo-sine VCO

Third Order Input Intercept Point (IIP3) obtained from QPSS and QPAC analysis is shown in figure 8. For sine wave, IIP3 is 6.01028 dBm & for Pseudo-sine wave IIP3 is 11.5961 dBm. IIP3 obtained by pseudo-sine wave is improved by around 5.58dB. The voltage conversion gain is obtained from PSS & PAC analysis. It is represented in figure 9, for sine wave shown by green dashed line maximum gain is 5.307 dB and for Pseudo-sine wave shown by red, maximum gain is 6.7121dB. The Noise Figure (NF_{DSB}) shown in figure 10, obtained from PSS & PNoise analysis for sine wave shown by green, minimum NF_{DSB} is 10.79 dB and for Pseudo-sine wave shown in red is 10.30dB. The layout of the complete design is shown in figure 11, the length and width of the design is 39.22 μ m and 33.80 μ m respectively. All measurements are taken 2λ away from the design. The total area consumed by the design is 1325.636 μ m² or 0.001325mm².



(a) For sine wave



(b) For Pseudo-sine wave

Fig. 8. IIP3 using QPSS and QPAC

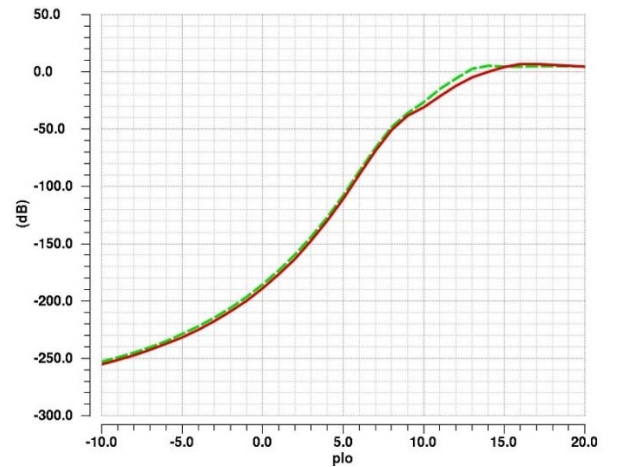


Fig. 9. Voltage Conversion Gain versus the LO Signal Power

Noise Figure (NF _{dsb})	10.79 dB	10.30 dB
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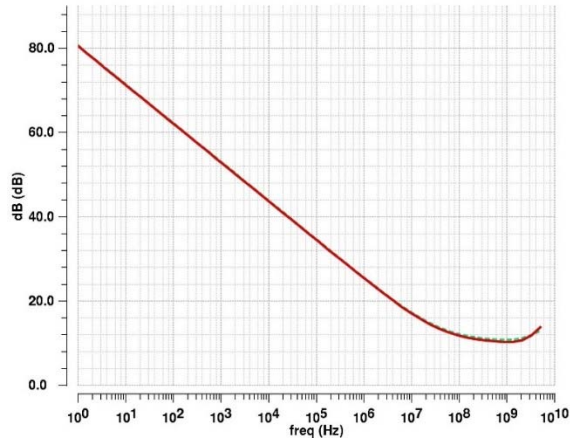


Fig. 10. Noise Figure DSB

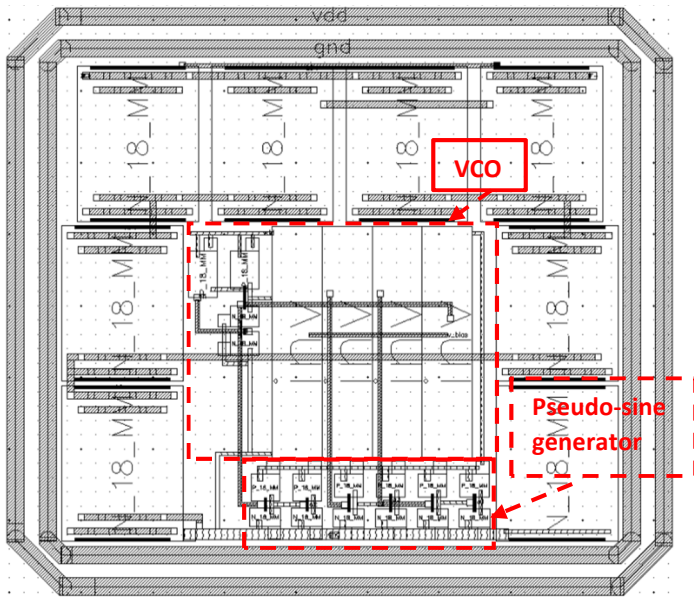


Fig. 11. Layout of complete design

The detailed comparison in UMC 180nm process is shown in Table II and the width of MOSFETs are specified in Table III.

TABLE II. Comparison between Sine and Pseudo-Sine wave

	Sine wave	Pseudo-sine wave
Voltage (V)	1.8 V	1.8 V
Current (I)	2.286 mA	2.286 mA
Power (P)	4.1148 mW	4.1148 mW
Conversion Gain (dB)	5.307 dB	6.712078 dB
3 rd order Input Intercept point (IIP3)	6.01028 dBm	11.5961 dBm

TABLE III. Width specification of pseudo-sine VCO

Component	Width/Length or Value
M1, M4, M7, M10	7μm/0.18μm
M2, M5, M8, M11	3μm/0.18μm
M3, M6, M9, M12	1μm/0.18μm
M13	0.8μm/0.18μm
M14	0.6μm/0.18μm
M15	1.8μm/0.18μm
M16	2.4μm/0.18μm
M17, M19, M25, M27	0.24μm/ 0.18μm
M18, M20, M26, M28	0.72μm/ 0.18μm
M21, M23	.034μm/0.18μm
M22, M24	1.02μm/0.18μm
R1, R3	1KΩ
R2	1.414KΩ

By definition of SFDR given in [2-3], the SFDR achieved by conventional LO is compared with pseudo-sine wave LO.

$$SFDR = \frac{2}{3} \cdot (IIP3 - P_n) \quad (17)$$

In (17), P_n is the noise floor. For 2.4GHz IoT applications, P_n is calculated in [1-2] as -111dBm. Using (17), the SFDR achieved by pseudo-sine LO is 81.72dB which is 3dB higher than the SFDR achieved by sine type LO applied to the Gilbert mixer shown in Fig. 7.

V. CONCLUSION

It is observed from the study that using Pseudo-Sine LO designed in UMC 180nm process when applied to the Gilbert mixer, it improves the IIP3 by 5.58dB over the IIP3 obtained by normal sine wave LO without increasing the power. Pseudo-sine wave rejects 3rd and 5th order harmonics completely by definition. However, due to parasitic effects, small amount of HD is present in the circuit. Hence, it can be concluded that using pseudo-sine instead of conventional sine wave LO, the dynamic range of the wireless device can be improved.

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